CIE AS PHYSICS (9702) • QUESTION PAPER

8.2 Diffraction

Syllabus Section: 8. Superposition **Paper Type:** Structured Questions (Paper 2) **Total:** 5 Questions (41 Marks)

Question 1 Easy [10 marks]

Part (a) [2 marks]

State what is meant by the *diffraction* of a wave.

Part (b) [2 marks]

State **one** property of a wave that changes when it is diffracted and **one** that remains the same.

Part (c) [2 marks]

Use words from Table 1.1 to complete the following sentence:

..... passing through ..... will produce a large diffraction effect.

Table 1.1

X–rays
a diffraction grating
red light
atoms
radio waves
mountains

Part (d) **[4 marks]**

Fig. 1.1 shows a child in a hallway looking for a toy which is making a noise and emitting light.

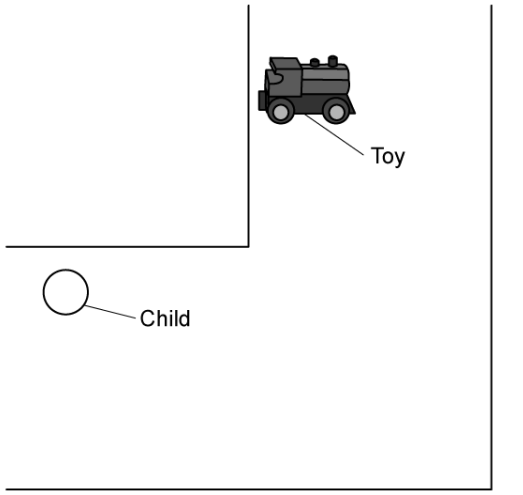


Fig. 1.1

The child can hear the sound of the toy, but cannot see the light coming from it.

Complete the following sentences by circling the correct word:

Sound waves emitted by the toy **diffract / refract** around the corner because the **amplitude / wavelength** is similar in size to the width of the hallway.

The **amplitude / wavelength** of light is much **larger / smaller** than that of the sound waves, so light is not **diffracted / refracted** around the corner.

Question 2 Easy [8 marks]

Part (a) [2 marks]

Wavefronts of a water wave pass through a gap in a harbour, as shown in Fig 1.1.

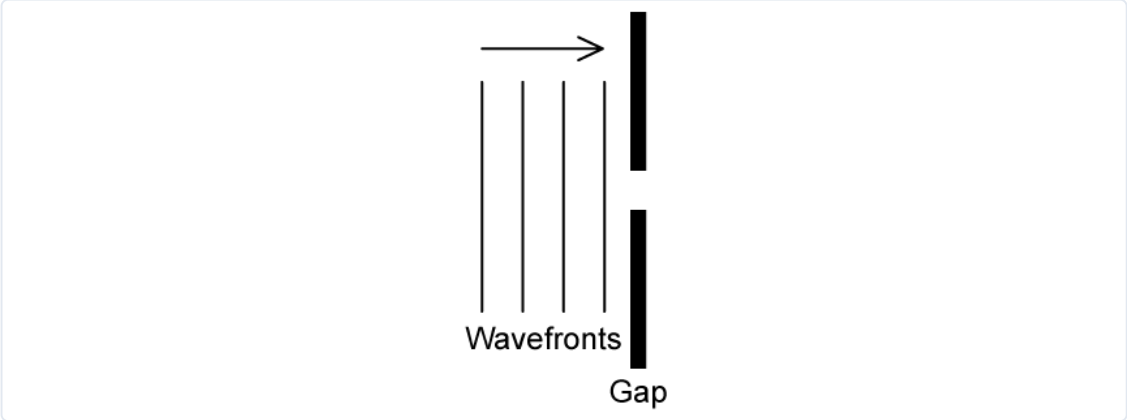


Fig. 1.1

Complete Fig 1.1 to show four wavefronts after they have emerged from the gap.

Part (b) [2 marks]

State **two** changes to Fig. 1.1 that would decrease the amount of diffraction of the water waves.

Part (c) [2 marks]

Fig. 1.2 shows a water wave meeting a barrier.

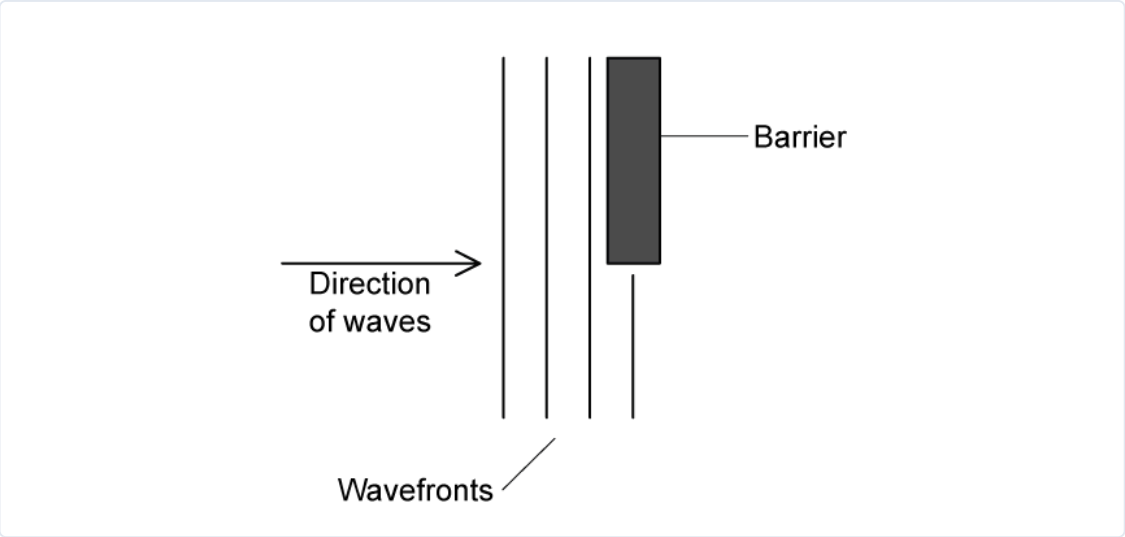


Fig 1.2

Complete Fig. 1.2 to show the wavefronts after they have passed the barrier.

Part (d) [2 marks]

Fig. 1.3 shows the wavefronts of a water wave moving from left to right in a harbour.

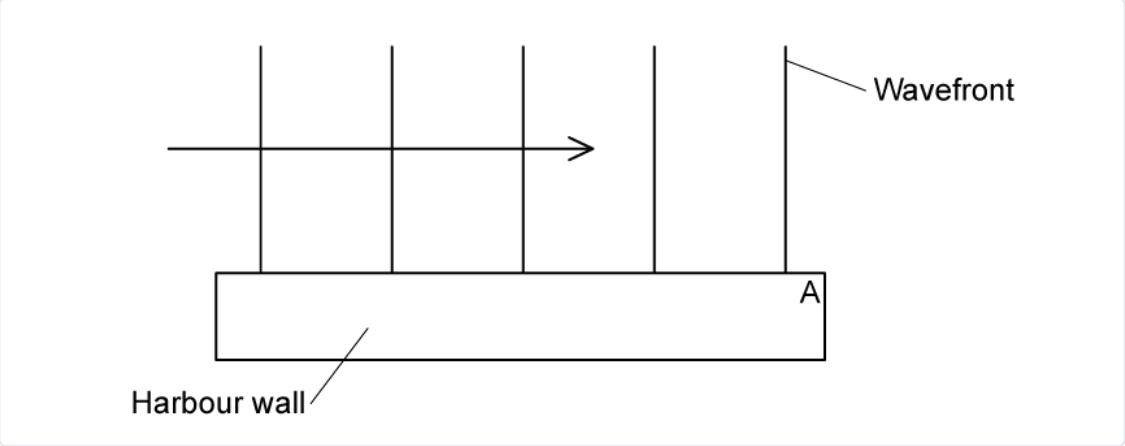


Fig. 1.3

Complete Fig. 1.3 to show the wavefronts as they pass to the right of point A.

Question 3 Medium [9 marks]

Part (a) [2 marks]

White light is passed through a single narrow slit and illuminates a screen.

Describe the pattern observed on the screen.

Part (b) [2 marks]

Blue light from a laser is now passed through a single narrow slit, as shown in Fig 1.1. A pattern of bright and dark regions can be observed on the screen which is placed several metres beyond the slit.

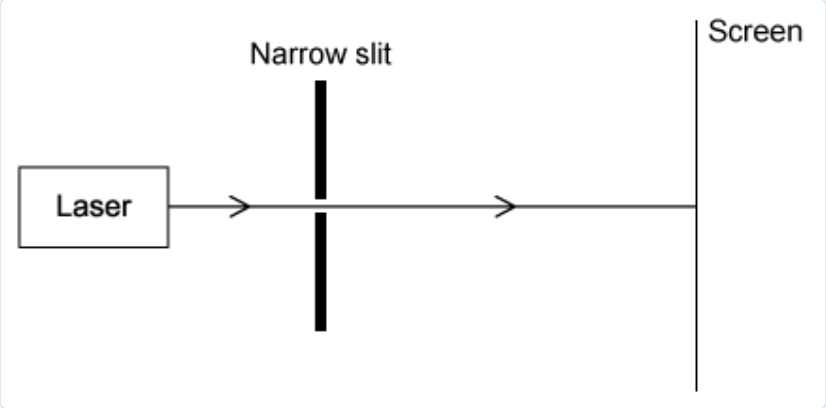


Fig. 1.1

Describe the effect on the diffraction pattern if the width of the narrow slit is decreased.

Part (c) [2 marks]

With the original slit width, state and explain the effect on the width of the fringes on the diffraction pattern if the blue light is replaced with a red light of the same intensity.

Part (d) [3 marks]

The intensity graph for the diffracted blue light is shown in Fig 1.2.

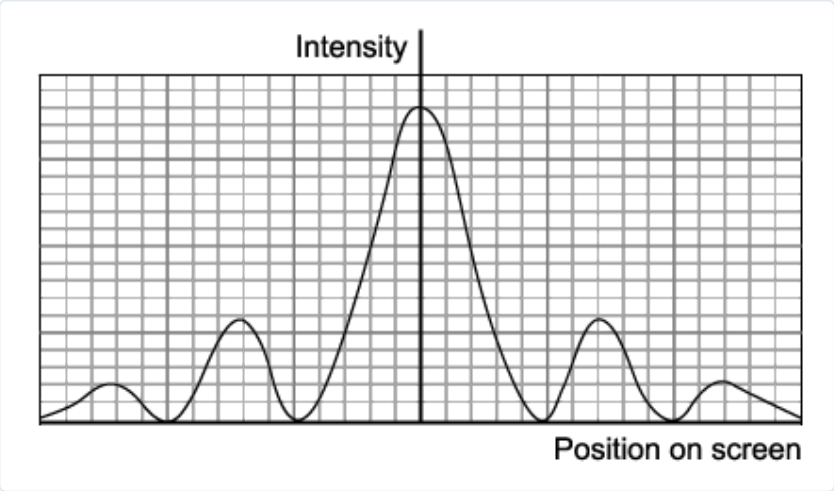


Fig. 1.2

On the axes shown in Fig 1.2, sketch the intensity graph for the laser emitting red light.

Question 4 Medium [6 marks]

Part (a) [2 marks]

A beam of monochromatic light is incident upon two slits. The distance between the slits is *d* as shown in Fig 1.1.

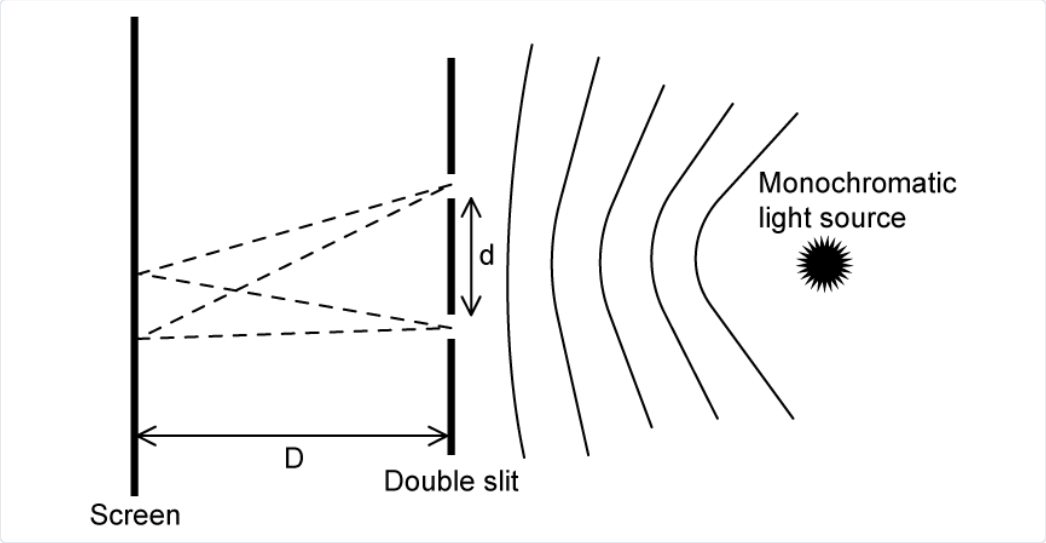


Fig 1.1

A series of bright and dark fringes appear on the screen. Explain how a bright fringe is formed.

Part (b) [2 marks]

If the wavelength of the incident light is halved and the distance between the slits is doubled, outline the effect on the separation of the fringes of the interference pattern.

Part (c) [2 marks]

One of the slits is covered so it emits no light.

Describe how this changes the pattern's appearance and the intensities observed on the screen.

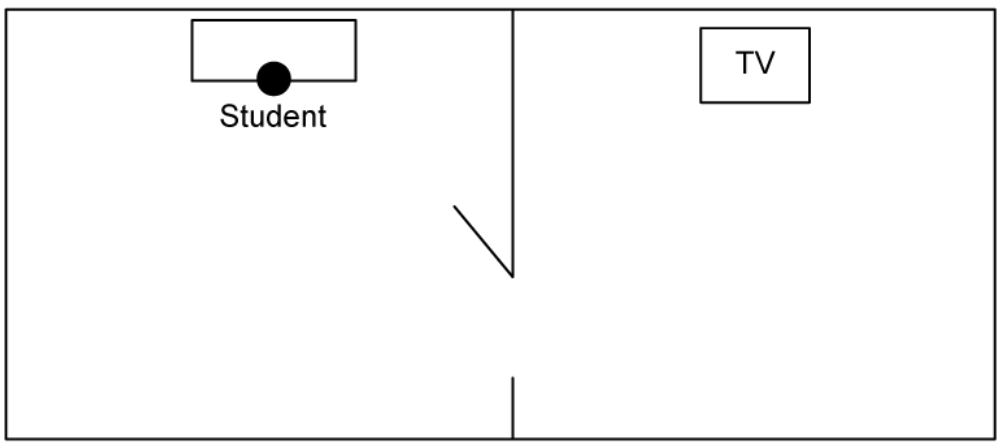
Question 5 Medium [8 marks]

Part (a) [1 mark]

Define diffraction.

Part (b) [4 marks]

The diagram shows a student in a room with an open door and a television in the next room.



The student can hear the sound of the television, but cannot see the picture coming from it.

Explain why this is the case.

Part (c) [3 marks]

Explain what would happen in terms of diffraction of the sound waves, if the student's room mate partially closed the door.